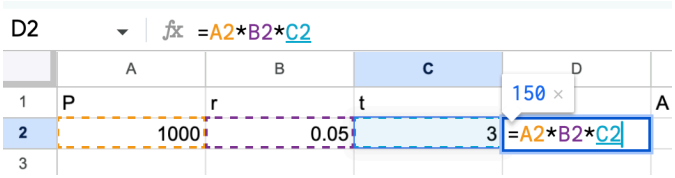
1. **Simple Interest** Suppose Liz borrows $1000 and agrees to pay it back in 3 years with 5% simple interest added each year.

   (a) How much **interest** will she pay? In addition to your answer, write down your computation, including the decimal values you would plug into a calculator/spreadsheet.

Compute

$I = P * r * t$

$= 1000 * 0.05 * 3$

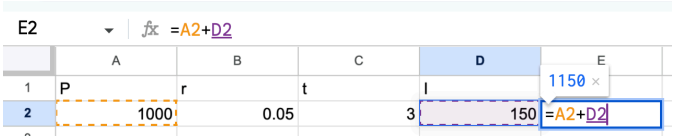


   (b) How much will she owe **in total** at the end of three years? Write down the computation and the answer.

Compute

$A = P + I$

$= 1000 + 1000(0.05)(3)$



2. **Annual Interest Rate (APR)** Annual interest rates are pro-rated by time.

   (a) Definition: An APR of r% paid n times per year means the interest rate is pro-rated:

each time interest is computed, you use an interest rate of $\frac{r}{n}$ as a decimal, not a percent!

   (b) So an APR of 5% paid **quarterly** means

every 3 months, you compute interest using a rate of $\frac{0.05}{4} = 0.0125$ (3 months = $\frac{1}{4}$ of a year). Every quarter, you compute interest using 1.25% as the rate.

3. Determine n for each word below.

word:	annually	semi-annually	quarterly	monthly	weekly	daily
n times per year	1	2	4	12	52	365

Compound Interest

4. What is **compound** interest and how is it different from **Simple** Interest?

You add the interest back into the principal (instead of putting the interest into your wallet). Next time, you compute interest on the combined amount!

5. Suppose \$1000 will be invested in an account that accrues **compound** interest at an APR of 3% compounded **monthly**. We will set up a spreadsheet to model this account assuming no other deposits or withdrawals.

   (a) Open a new spreadsheet and type column headers in cells `A1, B1, C1, D1, E1, F1`. (See below.)

   (b) In row 2, type the variables for reference. (hint: `A2` is r.)

   (c) In row 3, enter the **start** values for this specific problem. Assume time t is in ~~years~~ months!

   (d) In rows 4 and 5, enter the values/formulas ~~and~~ at the end of 1 month and 2 months.

   (e) Use the pull down feature to determine the values in the account at the end of 2 years. = 24 months

(see next page)

	A	B	C	D	E	F
1	Interest Rate	Times per Year	Principal	Time Elapsed	Interest	Total
2	r	n	P	t (months!)	I	A
3	0.03	12	1000	0		=C3
4				=D3+1	=see below...	=E4+F3
5				=D4+1		

Interest = \$A\$3/\$B\$3 * F2

Next interest = (previous total) * r/n

6. Write out the month-by-month computations and look for a pattern.

month 1: $A = 1000\left(1 + \frac{0.03}{12}\right)$ ← this is our "new P"!

month 2: $A = \underbrace{1000\left(1 + \frac{0.03}{12}\right)}_{\text{new P}}\left(1 + \frac{0.03}{12}\right) = 1000\left(1 + \frac{0.03}{12}\right)^2$

month 3: $A = 1000\left(1 + \frac{0.03}{12}\right)^2\left(1 + \frac{0.03}{12}\right) = 1000\left(1 + \frac{0.03}{12}\right)^3$

guess after 2 years: $A = 1000\left(1 + \frac{0.03}{12}\right)^{24}$ ...

2 years = 24 months = 12 · 2 ← n · t

7. **Compound Interest Formula:** A principal value of P dollars earning r% interest compounded n times per year for t years will result in a total of A dollars where

$$A = P\left(1 + \frac{r}{n}\right)^{n \cdot t}$$

8. **Effective Annual Interest Rate:**

Compute total interest earned over 1 year, divide by P, and convert to a %: $\frac{I}{P} = \frac{A-P}{P} = \frac{A}{P} - 1$

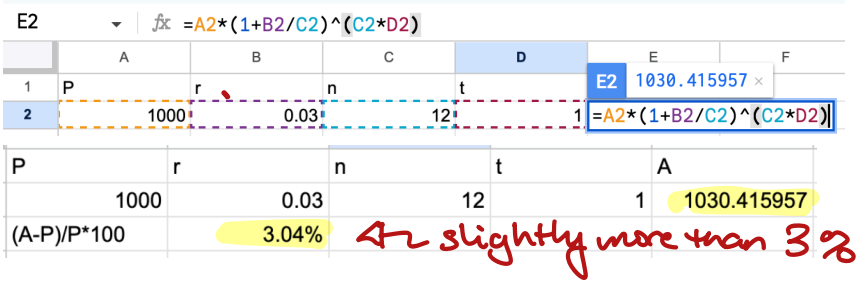
Lets you compare over different compounding periods!

9. What is the effective annual interest rate on \$1000 invested at a annual percentage rate of 3% compounded monthly?

one year = 12 months.

Compute $A = 1000\left(1 + \frac{0.03}{12}\right)^{1 \cdot 12}$

and then $\left(\frac{A-P}{P}\right)$



← slightly more than 3%!

Spreadsheet computations from #5

D31

	A	B	C	D	E	F
1	interest rate	times per year	principal	Time elapsed	interest	total
2	r	n	P	#months	I	A
3	0.03	12	1000	0		1000
4				1	$2.50	$1,002.50
5				2	$2.51	$1,005.01
6				3	$2.51	$1,007.52
7				4	$2.52	$1,010.04
8				5	$2.53	$1,012.56
9				6	$2.53	$1,015.09
10				7	$2.54	$1,017.63
11				8	$2.54	$1,020.18
12				9	$2.55	$1,022.73
13				10	$2.56	$1,025.28
14				11	$2.56	$1,027.85
15				12	$2.57	$1,030.42
16				13	$2.58	$1,032.99
17				14	$2.58	$1,035.57
18				15	$2.59	$1,038.16
19				16	$2.60	$1,040.76
20				17	$2.60	$1,043.36
21				18	$2.61	$1,045.97
22				19	$2.61	$1,048.58
23				20	$2.62	$1,051.21
24				21	$2.63	$1,053.83
25				22	$2.63	$1,056.47
26				23	$2.64	$1,059.11
27				24	$2.65	$1,061.76